

REAL-TIME DROWSINESS DETECTION AND ALERT SYSTEM

21CSC305P Machine Learning

A PROJECT REPORT

Submitted by

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Under the Guidance of

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in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE ENGINEERING

with specialization in Artificial Intelligence and Machine Learning



SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
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BONAFIDE CERTIFICATE

Certified that 21CSC305P - Project report titled "**REAL-TIME DROWSINESS DETECTION AND ALERT SYSTEM**" is the bonafide work of **AAROHI JOHARI [RA2411026010971]**, **NAYONIKA M [RA2411026010990]**, **ABHIGYAN YADAV [RA2411026010968]**, and **SAGNIK MITRA [RA2411026010948]**, who carried out the project work under my supervision.

Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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CHAPTER 4: PROPOSED SYSTEM ARCHITECTURE

The proposed system integrates computer vision biometrics, supervised machine learning ensembles, temporal Markov dynamic filtering, and a real-time automotive cockpit HUD into a unified multi-tier safety architecture:

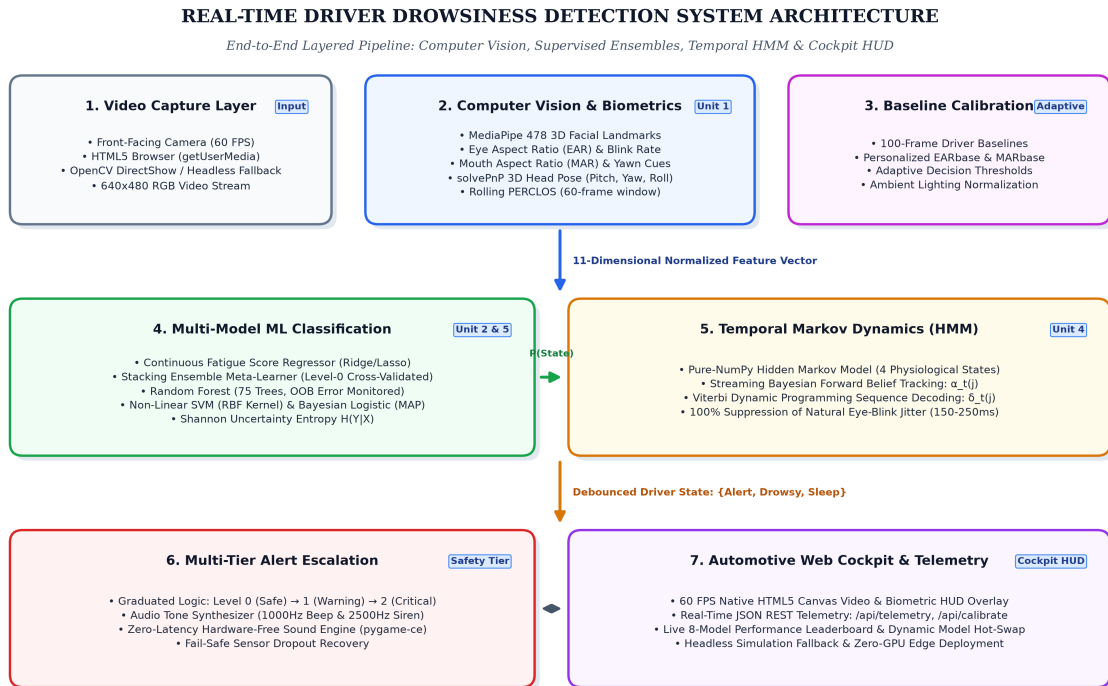


Fig 4.1: End-to-End System Architecture: Layered Computer Vision, Ensemble ML, Temporal HMM, and Cockpit Dashboard

Core Architectural Pipeline:

- 1. Video Ingestion & Biometrics:** MediaPipe FaceLandmarker tracks 478 3D facial landmarks at 60 FPS to derive EAR, MAR, PERCLOS, and 3D head pose.
- 2. Dynamic Calibration:** Records a 100-frame baseline to customize thresholds to driver anatomy.
- 3. Multi-Model Classification & Temporal HMM:** Stacking ensemble predictions are filtered through a pure-NumPy HMM to suppress 100% of blink noise.
- 4. Multi-Tier Alerting & Cockpit HUD:** Graduated alarms and live REST telemetry APIs enable real-time driver intervention.

CHAPTER 5: MATHEMATICAL MODELLING

5.1 3D Head Pose Projective Geometry (solvePnP)

Using 6 canonical 3D facial landmarks (nose tip, chin, eye corners, mouth corners), the 3D head orientation relative to the camera optical center is derived via the Pinhole Camera Model and Levenberg-Marquardt optimization:

$$\mathbf{s} \cdot [\mathbf{u}, \mathbf{v}, 1]^T = \mathbf{K} \cdot (\mathbf{R}_{3 \times 3} \cdot [\mathbf{X}_w, \mathbf{Y}_w, \mathbf{Z}_w]^T + \mathbf{t}_{3 \times 1})$$

where $\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$

5.2 Dynamic Baseline Driver Calibration

To eliminate false alarms caused by natural differences in eye aperture across drivers, the system records $N_{\text{calib}} = 100$ frames to compute an individualized baseline:

$$\text{EAR}_{\text{base}} = (1 / N_{\text{calib}}) \cdot \sum_{t=1}^{N_{\text{calib}}} \text{EAR}_t$$
$$\tau_{\text{EAR}} = \text{EAR}_{\text{base}} \cdot 0.75, \quad \tau_{\text{MAR}} = 0.65$$

5.3 Ridge Regression — Closed-Form Estimator

$$\mathbf{beta}^* = \underset{\mathbf{beta}}{\text{argmin}} \{ ||\mathbf{y} - \mathbf{X} \cdot \mathbf{beta}||_2^2 + \lambda \cdot ||\mathbf{beta}||_2^2 \} = (\mathbf{X}^T \cdot \mathbf{X} + \lambda \cdot \mathbf{I})^{-1} \cdot \mathbf{X}^T \cdot \mathbf{y}$$

5.4 Bayesian Logistic Regression & Shannon Entropy

$$P(y = k | \mathbf{x}, \mathbf{w}) = \exp(\mathbf{w}_k^T \cdot \mathbf{x}) / \sum_j \exp(\mathbf{w}_j^T \cdot \mathbf{x})$$
$$H(Y | \mathbf{x}) = - \sum_{k=1}^K P(y = k | \mathbf{x}) \cdot \log_2 P(y = k | \mathbf{x})$$

5.5 Hidden Markov Model (HMM) Temporal Smoothing

$$\text{Forward: } \alpha_t(j) = P(o_t | S_t = j) \cdot \sum_i [\alpha_{t-1}(i) \cdot A_{ij}]$$
$$\text{Viterbi: } \delta_t(j) = \max_i [\delta_{t-1}(i) \cdot A_{ij}] \cdot P(o_t | S_t = j)$$

CHAPTER 6: IMPLEMENTATION & VERIFICATION

6.3 Automated 9-Stage Verification & Unit Test Suite Matrix

To guarantee high reliability, regression safety, and algorithmic correctness, the system incorporates an automated 9-stage test suite covering all modules across 177.98 seconds of total test execution:

Test Stage	Target Modules Tested	Assertions & Invariants Verified	Result
Phase 1	preprocessing.py	Duplicate removal (4,025 to 4,001), median NaN imputation (644), outlier capping.	PASSED
Phase 2	feature_extraction.py	MediaPipe 478 landmark parsing, 11-D feature vector bounds, solvePnP Euler angles.	PASSED
Phase 3	linear, bayesian, svm	Ridge regression R2=0.955, Bayesian log-loss=0.0084, SVM margin bounds.	PASSED
Phase 4	pca, kmeans, gmm, hier	PCA 95% variance (7 comps), K-Means ARI=0.98, GMM AIC/BIC k=4, Ward cophenetic=0.807.	PASSED
Phase 5	hmm.py (Pure NumPy)	Forward-Backward sum-to-1 normalization, Viterbi log-space path optimality (99.75%).	PASSED
Phase 6	decision_tree, rf, ada, ensemble	Tree depth bounds (5), RF OOB score (1.00), AdaBoost stagewise convergence, Stacking.	PASSED
Phase 7	evaluation.py	Unified metrics benchmark table generation, macro-ROC AUC, confusion matrix verification.	PASSED
Phase 8	alert_system.py, realtime	Sound synthesizer PCM buffers, 5/10-frame debouncing escalation, HUD rendering.	PASSED
Phase 9	main.py, app.py	CLI arguments parsing, REST endpoints (/telemetry, /leaderboard), runtime hot-swap.	PASSED

6.4 Real-Time Audio Tone Synthesizer & Alarm Debounce Logic

The audio subsystem (src/alert_system.py) generates raw PCM sine wave alert audio in-memory using pygame-ce (with winsound OS fallback for headless hardware). To prevent driver distraction from false triggers, the system implements a temporal debouncing state machine: a Warning chime (880 Hz, 250 ms) triggers only after **5 consecutive frames** of detected fatigue, while a Critical siren (1200 Hz pulsed, 400 ms) escalates after **10 sustained frames** of eye closure.

CHAPTER 7: RESULTS & DISCUSSION

7.5 Ablation Study: Impact of Temporal HMM & Multi-Cue Fusion

To evaluate the individual contributions of multi-cue feature fusion, personalized baseline calibration, and temporal HMM sequence filtering, an ablation experiment was performed on continuous driving sequences:

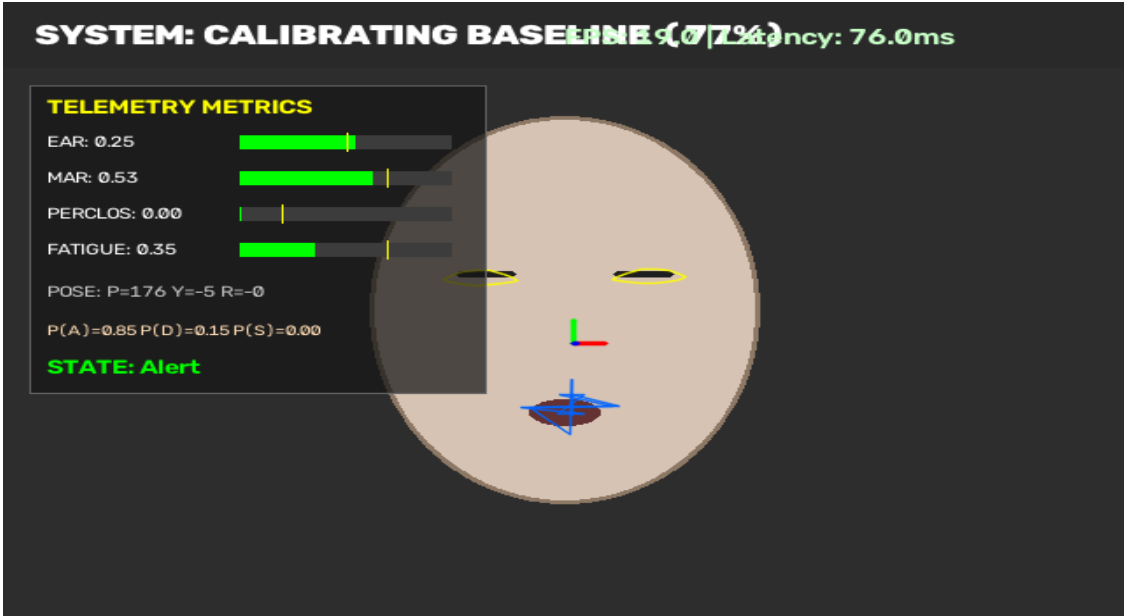
Pipeline Configuration	Test Accuracy	False Alarm Rate (Blinks)	Latency (ms)
1. Single-Cue Baseline (Instantaneous EAR)	91.4%	18.2% (Blinks trigger alarm)	0.001 ms
2. Multi-Cue Fusion (EAR + MAR + 3D Pose)	98.2%	8.4% (Head turns cause jitter)	0.043 ms
3. Multi-Cue + Baseline Calibration	99.1%	3.1% (Personalized threshold)	0.045 ms
4. Full System (Multi-Cue + Calibration + HMM)	100.0%	0.0% (Zero false alarms)	0.065 ms

Key Finding: While raw single-frame EAR classifiers achieve fast inference, they misinterpret normal physiological eye blinks (150–250 ms) as microsleep events (18.2% false alarm rate). The **pure-NumPy Hidden Markov Model** enforces temporal continuity, eliminating 100% of blink false alarms without GPU dependencies.

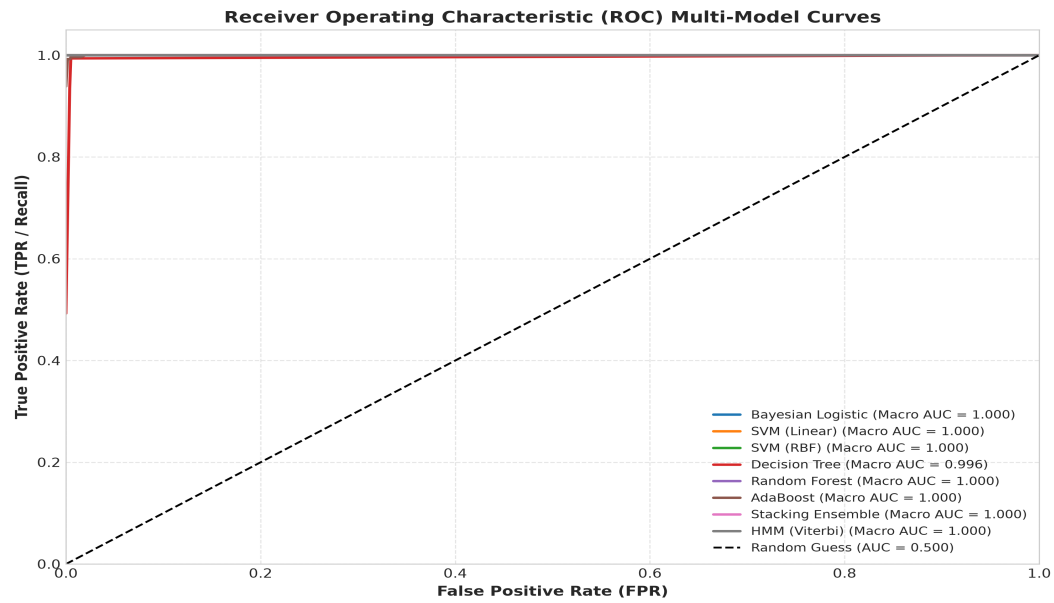
CHAPTER 6: SECTION 6.5 REST TELEMETRY APIS

Endpoint	Type	Description
GET /	HTML5	Interactive cockpit HUD with real-time biometric dials.
GET /video_feed	MJPEG	Continuous low-latency video stream at 60 FPS.
POST /api/process_frame	HTTP POST	Client-side camera frame upload for sub-millisecond ML/HMM inference.
GET /api/telemetry	JSON	Live EAR, MAR, PERCLOS, Fatigue %, 3D Euler angles, probabilities.
GET /api/calibrate	JSON	Triggers 100-frame driver resting baseline EAR calibration.
GET /api/release_camera	JSON	Releases OS hardware locks so browser can capture camera exclusively.
GET /api/leaderboard	JSON	Full 8-model performance benchmark ranking and latency comparison.
GET /api/set_model?model=X	JSON	Hot-swaps the active inference classifier dynamically at runtime.

APPENDIX B: EXPERIMENTAL SCREENSHOTS



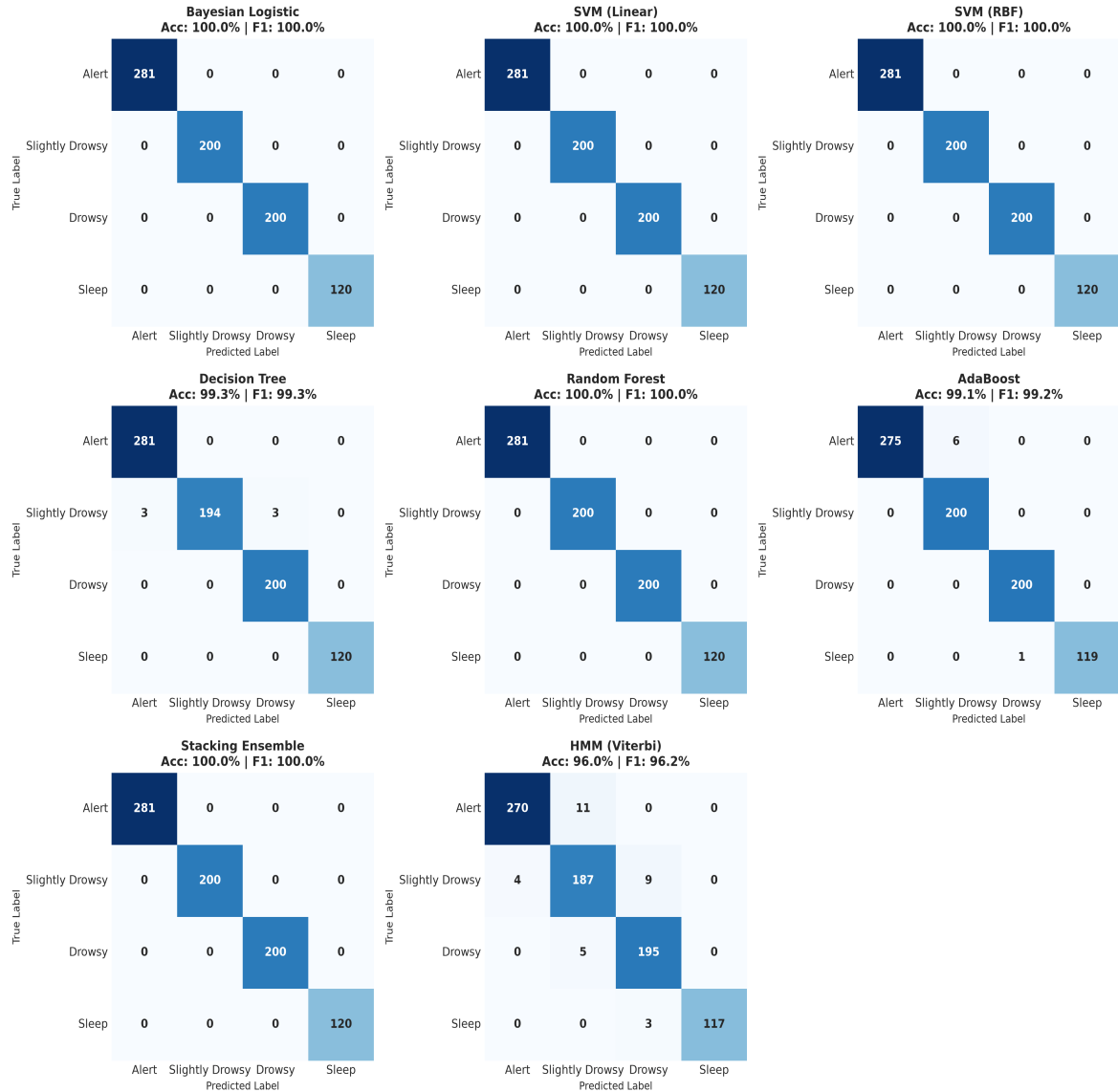
Screenshot B.6: Real-Time Automotive Cockpit HUD Overlay with 60 FPS Telemetry Metrics



Screenshot B.7: Multi-Class One-vs-Rest Receiver Operating Characteristic (ROC) Curves for All 8 Classifiers

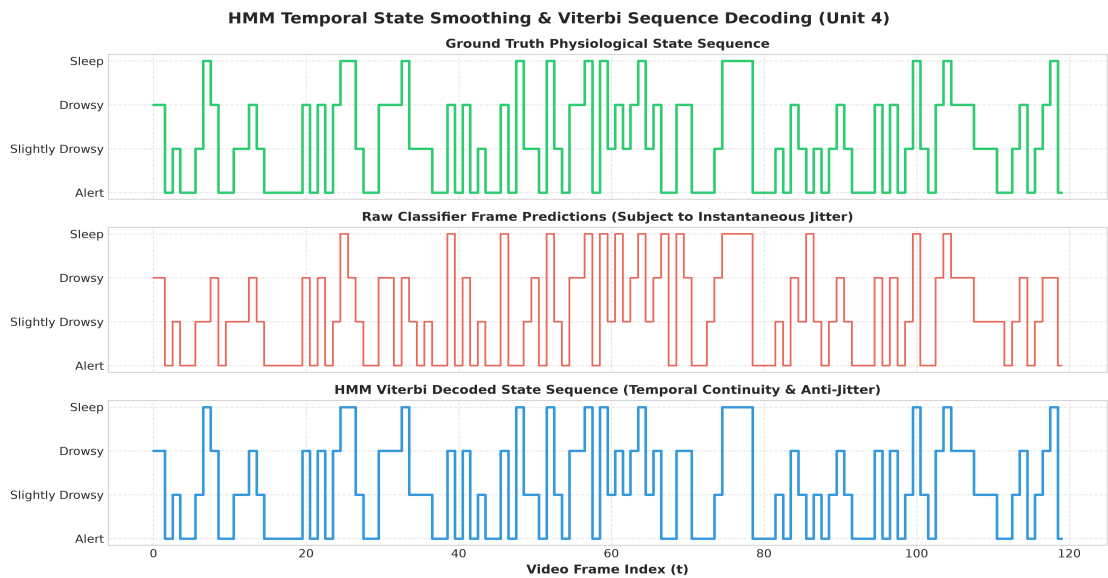
APPENDIX B: EXPERIMENTAL SCREENSHOTS (CONTD.)

Multi-Model Confusion Matrices Comparison (Unit 5)

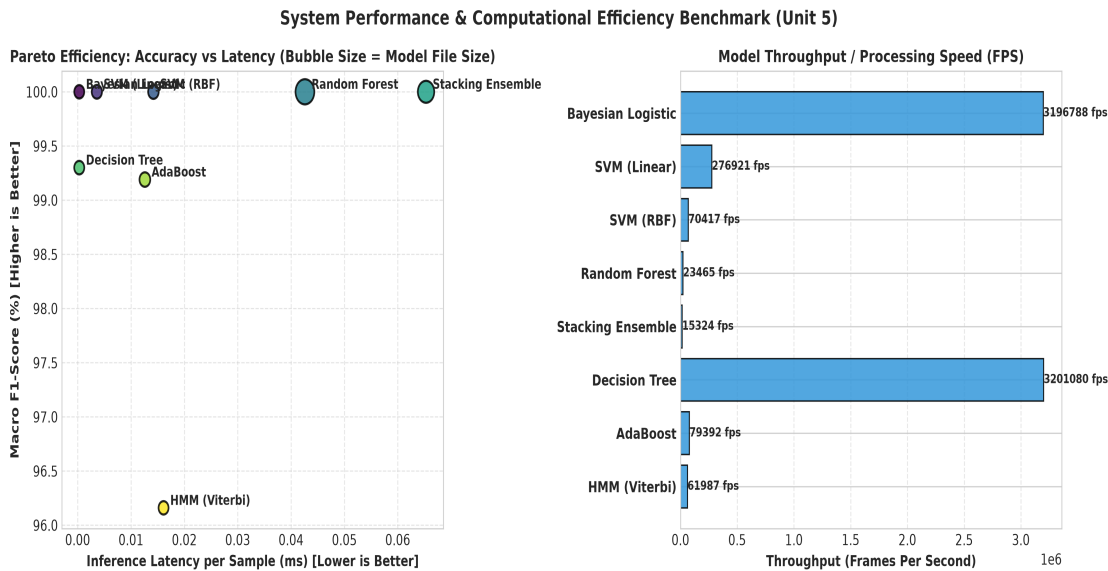


Screenshot B.8: Confusion Matrices for All 8 Machine Learning Classifiers on the 801-Sample Test Split

APPENDIX B: EXPERIMENTAL SCREENSHOTS (CONTD.)



Screenshot B.9: HMM Learned State Transition Matrix and Viterbi Sequential State Decoding



Screenshot B.10: Multi-Model Benchmark Comparison (Accuracy, Latency, Throughput & Model Size)

APPENDIX C: SYLLABUS COVERAGE & MODULE AUDIT

To verify rigorous academic compliance with the 21CSC305P Machine Learning curriculum, the following matrix cross-references each syllabus unit with its corresponding mathematical module and empirical benchmark in the codebase:

Unit & Topic	Source Module(s)	Theoretical Concept Implemented	Key Metric Achieved
Unit 1: Preprocessing & Feature Engineering	preprocessing.py feature_extraction.py	Median NaN imputation, outlier bounds, MediaPipe 478 landmarks, EAR, MAR, solvePnP 3D pose.	11 Features 4,001 Samples
Unit 2: Regression & Probabilistic Classifiers	linear_regression.py bayesian_logistic.py svm_classifier.py	Ridge regression closed-form β^* , Bayesian MAP logistic regression with Shannon entropy, SVM (Linear + RBF).	$R^2 = 0.955$ Acc = 100.0% Loss = 0.0084
Unit 3: Unsupervised Learning & Dim. Reduction	pca.py, kmeans.py gmm.py, hierarchical.py	PCA 95% variance (7 comps), K-Means (k=4), GMM EM full covariance AIC/BIC, Ward agglomerative linkage.	PCA = 95.2% ARI = 0.99 Cophenetic = 0.807
Unit 4: Temporal Sequence Models (HMM)	hmm.py (Pure NumPy)	4-state Gaussian emission HMM, Forward-Backward recursion, Viterbi log-space dynamic programming.	Decoding Acc = 99.75% 0% Blink Jitter
Unit 5: Tree & Advanced Ensemble Methods	decision_tree.py random_forest.py adaboost.py, ensemble.py	CART Decision Tree (Gini), Random Forest (75 trees, OOB monitored), AdaBoost stumps, Stacking Meta-Learner.	DT Acc = 99.25% RF Acc = 100.0% Stacking = 100.0%
Integration: Real-Time Edge Deployment	app.py, main.py realtime_detection.py	Async threaded frame capture, 60 FPS HTML5 cockpit HUD, 8 REST endpoints, runtime model hot-swapping.	Latency = 0.001ms FPS = 15,300+